Probabilistic Combinatorics

Sheet 4

May 13, 2024

Problem 1. There is a chess tournament with 40 players with 80 games played, where every two players face each other at most once. Proof:

(i) There are at least 8 players, no two of which played each other.

(ii) For real numbers $a_1, \ldots, a_r > -1$ holds

$$\frac{1}{a_1+1} + \ldots + \frac{1}{a_r+1} \geq \frac{r^2}{(a_1+1)+\ldots+(a_r+1)}$$

Proof.
Let P be the set of players in the chess tournament. Moreover, let the players in P uniquely labelled by numbers in $[40]$, which means $P = [40]$. Next, let the bijective function $f : P \to [40]$ be a ranking of players in the tournament. The idea to solve this problem is to determine whether there exists a ranking among all possible rankings where it is possible to pick at least 8 players who only played against players with lower rank. If such ranking exists, then the problem is solved since no two of these players played each other.

Let $(\Omega, \mathbb{P})$ be the discrete probability space where the sample space Ω is the set of all rankings, or bijective functions $f : P \to [40]$, while the probability measure $\mathbb{P}(\omega) = \frac{1}{40!}$ for any $\omega \in \Omega$. Next, we define the random variable $X : \Omega \to \mathbb{N}$ that assigns to any ranking $\omega \in \Omega$ the number of players in such ranking that only played against players with lower rank. For any player $p \in P$, let the random variable $X_p : \Omega \to \mathbb{N}$ defined by the following rule for any $\omega \in \Omega$

$$X_p(w) = \begin{cases} 1 \text{ if } p \text{ only played against players with lower rank} \\ 0 \text{ otherwise} \end{cases}$$

Next, we define the indicator variable $X_p = 1_{A_p}$ of event

$$A_p = \{\omega \in \Omega \mid \ p \text{ only played against players with lower rank}\}$$

then $\mathbb{E}(X_p) = \mathbb{E}(1_{A_p}) = \mathbb{P}(A_p)$. In order to calculate $\mathbb{P}(A_p)$, we must firstly know $|A_p|$, which can be calculated as follows. First of all, let a_p be the number of players that p played against. In order to rank p and the players that p played against, we pick $a_p + 1$ unique numbers from $[40]$, and there are $\binom{40}{a_p+1}$ ways to do so. In each way, we must use the largest number to rank p, and other numbers to rank the players that p played against. Moreover, the rank between the players that p played against is not important at all, this implies that there are $a_p!$ ways to rank the players that p played against. Therefore, there are $a_p!$ ways to rank p and the players that p played against for each subset of $[40]$ of size $a_p + 1$. Furthermore, for each such subset and each way to rank p and the players that p played against in a way that p only played against players with lower rank, the players that are not p or the ones that p played against can be ranked arbitrarily using $40-(a_p+1)$ unique numbers left, and there are $(40 - a_p - 1)!$ ways to do so. Therefore,

$$|A_p| = \binom{40}{a_p+1} a_p!(40 - a_p - 1)!$$

Hence,

$$\begin{aligned}
\mathbb{P}(A_p) &= \frac{|A_p|}{40!} \\
&= \frac{\binom{40}{a_p+1} a_p!(40-a_p-1)!}{40!} \\
&= \frac{40!}{(a_p+1)!(40-a_p-1)!} \frac{a_p!(40-a_p-1)!}{40!} \\
&= \frac{a_p!}{(a_p+1)a_p!} \\
&= \frac{1}{a_p+1}
\end{aligned}$$

Note that, $X = \sum_{p \in P} X_p$. By linearity of expectation and $P = [40]$, we have

$$\begin{aligned}
\mathbb{E}(X) &= \sum_{p \in P} \mathbb{E}(X_p) = \sum_{p \in P} \mathbb{P}(A_p) \\
&= \frac{1}{a_1+1} + \ldots + \frac{1}{a_{40}+1}
\end{aligned}$$

It follows from (ii) that

$$\begin{aligned}
&\mathbb{E}(X) \geq \frac{40^2}{(a_1+1) + \ldots + (a_{40}+1)} \\
\Leftrightarrow\ &\mathbb{E}(X) \geq \frac{40^2}{(a_1 + \ldots + a_{40}) + 40}
\end{aligned}$$

For two distinct players $p_1, p_2 \in P$ that played each other, p_2 and p_1 is counted in a_{p_1} and a_{p_2}, respectively. That is, the result of $a_1 + \ldots + a_{40}$ is actually the number of games in the chess tournament multiplied by 2. Therefore, the inequality above is equivalent to

$$\begin{aligned}
&\mathbb{E}(X) \geq \frac{40^2}{160+40} \\
\Leftrightarrow\ &\mathbb{E}(X) \geq 8
\end{aligned}$$

This implies that there must exist a ranking of players in the chess tournament where it is possible to pick at least 8 players that only played against players with lower rank. Due to the logic provided at the beginning of the proof, this implies that there are at least 8 players in the chess tournament that no two of which played each other, and the proof for (i) is finished.

In order to prove (ii), we want to use Cauchy-Schwarz inequality. To be more specific, we have

$$\begin{aligned}
&\left[\sum_{i=1}^{r} \left(\frac{1}{\sqrt{a_i+1}}\right)^2\right] \left[\sum_{i=1}^{r} \left(\sqrt{a_i+1}\right)^2\right] \geq \left[\sum_{i=1}^{r} \left(\frac{1}{\sqrt{a_i+1}}\right) \left(\sqrt{a_i+1})\right)\right]^2 \\
\Leftrightarrow\ &\left(\sum_{i=1}^{r} \frac{1}{a_i+1}\right) \left(\sum_{i=1}^{r} a_i + 1\right) \geq \left(\sum_{i=1}^{r} i\right)^2 \\
\Leftrightarrow\ &\left(\sum_{i=1}^{r} \frac{1}{a_i+1}\right) \left(\sum_{i=1}^{r} a_i + 1\right) \geq r^2 \\
\Leftrightarrow\ &\left(\sum_{i=1}^{r} \frac{1}{a_i+1}\right) \geq \frac{r^2}{(\sum_{i=1}^{r} a_i + 1)} \\
\Leftrightarrow\ &\frac{1}{a_1+1} + \ldots + \frac{1}{a_r+1} \geq \frac{r^2}{(a_1+1) + \ldots + (a_r+1)}
\end{aligned}$$

and the proof for (ii) is finished. □

Problem 2. A subset $\Sigma \subseteq [n]$ is called additively free, if for every two subsets $A, B \subseteq \Sigma$:

$$\sum_{i \in A} i = \sum_{i \in B} i \Rightarrow A = B$$

(i) Find an additively free subset $\Sigma \subseteq [n]$ such that $|\Sigma| = \lfloor \log_2(n) \rfloor + 1$

(ii) Use the Chebyshev's inequality to show: There is an upper bound for the cardinality of an additively free subset $\Sigma \subseteq [n]$ given by

$$|\Sigma| \leq \log_2(n) + \log_2 \log_2(n) + 1$$

Proof.
No idea. □

Problem 3. Consider the Erdős-Renyi model $G(n, p_n)$ for random generating graphs. Let X_1 be the number of isolated vertices in $G(n, p_n)$. Proof: If $p_n = \frac{c_n \log(n)}{n}$ for some $c_n \gg 1$, then

$$\mathbb{P}(X_1 > 0) = o(1)$$

Proof.
Recall that $G(n, p_n)$ is the discrete probability space $(\Omega_n, \mathbb{P})$ where the sample space Ω_n is the set of all undirected graphs $G = ([n], E)$ while the probability measure $P(G) = p_n^{|E|}(1 - p_n)^{\binom{n}{2} - |E|}$ for any $G \in \Omega_n$.

For easier interpretation, let $X_n : \Omega_n \to \mathbb{N}$ be the random variable that assigns to any graph $G \in \Omega_n$ the number of isolated vertices in G, and thus $X_n = X_1$. That is, the original problem turns into the problem of showing that

$$\lim_{n \to \infty} \mathbb{P}(X_n > 0) = 0 \tag{3.1}$$

For a graph $G \in \Omega_n$ and an arbitrary vertex numbered i in G, let the random variable $X_n^{(i)} : \Omega_n \to \mathbb{N}$ defined by the following rule

$$X_n^{(i)}(G) = \begin{cases} 1 \text{ if the vertex numbered } i \text{ is isolated in } G \\ 0 \text{ otherwise} \end{cases}$$

Next, we define the indicator variable $X_n^{(i)} = 1_{A_n^{(i)}}$ of event

$$A_n^{(i)} = \{G = ([n], E) \mid \text{ the vertex numbered } i \text{ is isolated in } G\}$$

then $\mathbb{E}(X_n^{(i)}) = \mathbb{E}(1_{A_n^{(i)}}) = \mathbb{P}(A_n^{(i)})$. For an arbitrary vertex numbered i to be isolated in a graph $G \in \Omega_n$, there must be no edges between this vertex and any other vertices in G. Moreover, there are $n - 1$ possible edges between this vertex and any other vertices in G, and the probability of choosing a graph $G \in \Omega_n$ that does not have a specific edge is $1 - p_n$; hence,

$$\mathbb{P}(A_n^{(i)}) = (1 - p_n)^{n-1} \tag{3.2}$$

Note that, $X = \sum_{i=1}^n X_n^{(i)}$. By linearity of expectation, we have

$$\begin{aligned} \mathbb{E}(X) &= \sum_{i=1}^n \mathbb{E}(X_n^{(i)}) \\ &= \sum_{i=1}^n \mathbb{P}(A_n^{(i)}) \\ &= n(1 - p_n)^{n-1} \end{aligned}$$

Now, we want to show that $\lim_{n\to\infty} \mathbb{E}(X_n) = 0$. From the equation of $\mathbb{E}(X_n)$ given above, as n goes to infinity, we recognise that $n(1-p_n)^{n-1}$ is no different from $n(1-p_n)^n$; hence,

$$\lim_{n\to\infty} \mathbb{E}(X_n) = \lim_{n\to\infty} n(1-p_n)^n$$

Recall that $\left(1-\frac{1}{x}\right)^{-x} > e$ and $p_n = \frac{c_n \log(n)}{n}$, we have

$$\begin{aligned}(1-p_n)^n &= \left[\left(1-\frac{1}{\frac{n}{c_n\log(n)}}\right)^{-\frac{n}{c_n\log(n)}}\right]^{-c_n\log(n)} \\ &\leq e^{-c_n\log(n)} \\ &\leq n^{-c_n}\end{aligned}$$

Therefore, the equation (3.2) is equivalent to

$$\lim_{n\to\infty} \mathbb{E}(X_n) \leq \lim_{n\to\infty} n^{1-c_n} \tag{3.3}$$

Recall that $c_n \gg 1$, we have

$$\lim_{n\to\infty} \frac{1}{c_n} = 0 \Leftrightarrow \lim_{n\to\infty} c_n = \infty \tag{3.4}$$

Due to the equation (3.4), the equation (3.3) is equivalent to

$$\lim_{n\to\infty} \mathbb{E}(X_n) \leq 0 \tag{3.5}$$

It is worth to note that, X_n only takes non-negative integers. This means that the lower bound of $\mathbb{E}(X_n)$ is 0. Combine this with the equation (3.5), we have

$$\lim_{n\to\infty} \mathbb{E}(X_n) = 0$$

Hence, $\mathbb{E}(X_n) = o(1)$. It follows from the Corollary 6.3 in the lecture that the equation (3.1) is satisfied, and the proof is finished. □

Problem 4. In the Erdős-Renyi model $G(n,p_n)$ let $p_n = \frac{1}{n}$. Show: If n is big enough, there is a constant $c > 0$, such that the expected number of vertices of degree ≥ 3 (i.e. the vertex lies in ≥ 3 edges), is at least cn.

Proof.
Recall that $G(n,p_n)$ is the discrete probability space $(\Omega_n, \mathbb{P})$ where the sample space Ω_n is the of all undirected graphs $G = ([n], E)$ while the probability measure $\mathbb{P}(G) = p_n^{|E|}(1-p_n)^{\binom{n}{2}-|E|}$ for any $G \in \Omega_n$.

Let $X_n : \Omega_n \to \mathbb{N}$ be the random variable that assigns to any graph $G \in \Omega_n$ the number of vertices of degree ≥ 3 in G. For a graph $G \in \Omega_n$ and an arbitrary vertex numbered i in G, let the random variable $X_n^{(i)} : \Omega_n \to \mathbb{N}$ defined by the following rule

$$X_n^{(i)}(G) = \begin{cases} 1 \text{ if degree of the vertex numbered } i \text{ in } G \text{ is } \geq 3 \\ 0 \text{ otherwise} \end{cases}$$

Next, we define the indicator variable $X_n^{(i)} = 1_{A_n^{(i)}}$ of event

$$A_n^{(i)} = \{G = ([n], E) \mid \text{ degree of the vertex numbered } i \text{ in } G \text{ is } \geq 3\}$$

then $\mathbb{E}(X_n^{(i)}) = \mathbb{E}(1_{A_n^{(i)}}) = \mathbb{P}(A_n^{(i)})$. For any vertex numbered i in G, as the set $A_n^{(i)}$ is vague, its probability can be calculated as follows

$$\begin{aligned}\mathbb{P}(A_n^{(i)}) &= 1 - \mathbb{P}(\{G = ([n], E) \mid \text{ degree of the vertex numbered } i \text{ in } G \text{ is } < 3\}) \\ &= 1 - (\mathbb{P}(X_n = 0) + \mathbb{P}(X_n = 1) + \mathbb{P}(X_n = 2)) \end{aligned} \tag{4.1}$$

The probability of the random variable X_n when its value is $0, 1, 2$ can be calculated as follows:

- In order to calculate $\mathbb{P}(X_n = 0)$, we want to find the probability of the following set

  $$\mathbb{P}(\{G = ([n], E) \mid \text{ degree of the vertex numbered } i \text{ in } G \text{ is } 0\})$$

  The set in such probability is the set of graphs $G \in \Omega_n$ where none of $n-1$ possible edges between the vertex numbered i and any other vertices in G can exist. Moreover, the probability of choosing a graph $G \in \Omega_n$ that does not have a specific edge is $1 - p_n$; hence,

  $$\mathbb{P}(X_n = 0) = (1 - p_n)^{n-1} \tag{4.2}$$

- In order to calculate $\mathbb{P}(X_n = 1)$, we want to find the probability of the following set

  $$\mathbb{P}(\{G = ([n], E) \mid \text{ degree of the vertex numbered } i \text{ in } G \text{ is } 1\})$$

  The set in such probability is the set of graphs $G \in \Omega_n$ where there is only one edge between the vertex numbered i and another vertex in G. There are $\binom{n-1}{1}$ ways to select a vertex in G that are not numbered i to be the one to which we draw an edge from the vertex numbered i. In each selection, the possibility to choose a graph $G \in \Omega_n$ that has a specific edge between selected vertex and the vertex numbered i in G is p_n. Moreover, in each selection, none of $n-2$ possible edges between the vertex numbered i and any other vertices in G that are not selected can exist. Therefore,

  $$\mathbb{P}(X_n = 1) = (n-1)p_n(1 - p_n)^{n-2} \tag{4.3}$$

- Based on the idea for calculating $\mathbb{P}(X_n = 1)$ described above, $\mathbb{P}(X_n = 2)$ can be calculated in the similar way; hence, we have

  $$\mathbb{P}(X_n = 2) = \binom{n-1}{2} p_n^2 (1 - p_n)^{n-3} \tag{4.4}$$

Based on the following equations: $(4.2), (4.3), (4.4)$, the equation (4.1) is equivalent to

$$\mathbb{P}(A_n^{(i)}) = 1 - (1 - p_n)^{n-1} - (n-1)p_n(1 - p_n)^{n-2} - \binom{n-1}{2} p_n^2 (1 - p_n)^{n-3} \tag{4.5}$$

Note that $X = \sum_{i=1}^{n} X_n^{(i)}$. By linearity of expectation, we have

$$\begin{aligned}\mathbb{E}(X) &= \sum_{i=1}^{n} \mathbb{E}(X_n^{(i)}) \\ &= \sum_{i=1}^{n} \mathbb{P}(A_n^{(i)}) \\ &= n\mathbb{P}(A_n^{(i)}) \end{aligned}$$

That is, when n is large enough and we choose a constant c such that $0 < c \leq \mathbb{P}(A_n^{(i)})$, whose value is described by the equation (4.5), then

$$\mathbb{E}(X) \geq cn$$

and the proof is finished. □